

Design and Implementation of Smart Monitoring System for Home Security and Automation

PRESENTED BY:

MABASA FANELO

MAKAU BRIDGET

MDLOVU SOBONGANKOSINI

RAKHUNWANA MAAMBELE

MUZI MNISI



INTRODUCTION & BACKGROUND

- Smart homes improve security and convenience
- Uses IoT technology
- Automated monitoring and control
- Need for intelligent monitoring
- Integration of sensors and cameras

PROBLEM STATEMENT

- Traditional home security systems are inefficient and lack automation.
- They do not provide real-time monitoring, remote access, or intelligent control.
- Homeowners and families who require better security, convenience, and safety.
- Residential homes and small smart home environments.
- Especially when users are away from home or during emergencies and intrusions.



SOLUTION

- Smart home automation and security system
- Motion detection and alarm activation
- Automatic door operation
- Intelligent outdoor lighting
- Temperature monitoring and control
- Camera surveillance system
- Remote monitoring through web application

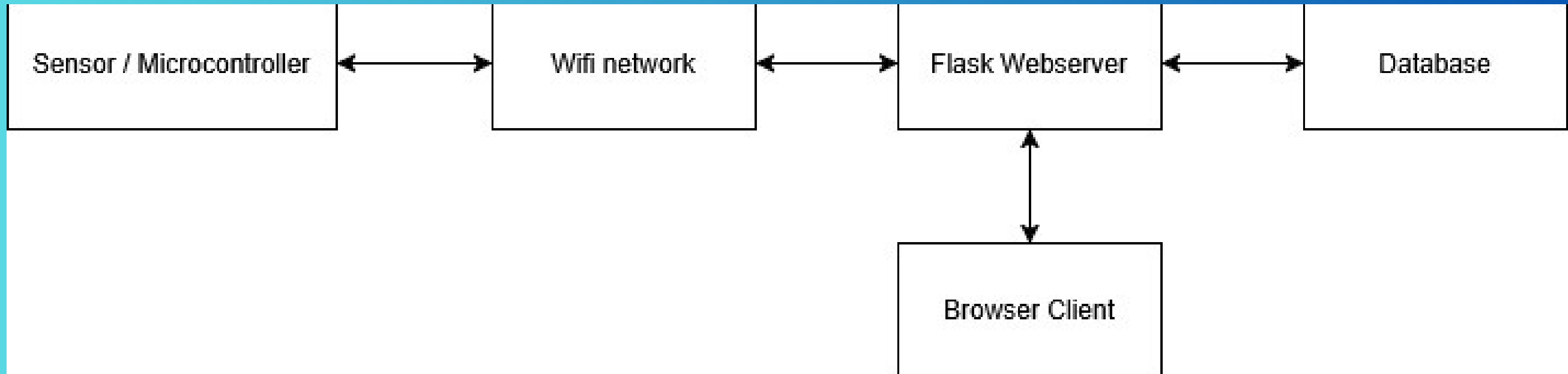


SOFTWARE

- C++ Arduino Language
- Wi-Fi Network
- Client/Server Communication
- Flask Web Application
- MySQL Database
- Live Camera Streaming
- Graph Generation
- GUI Client Page



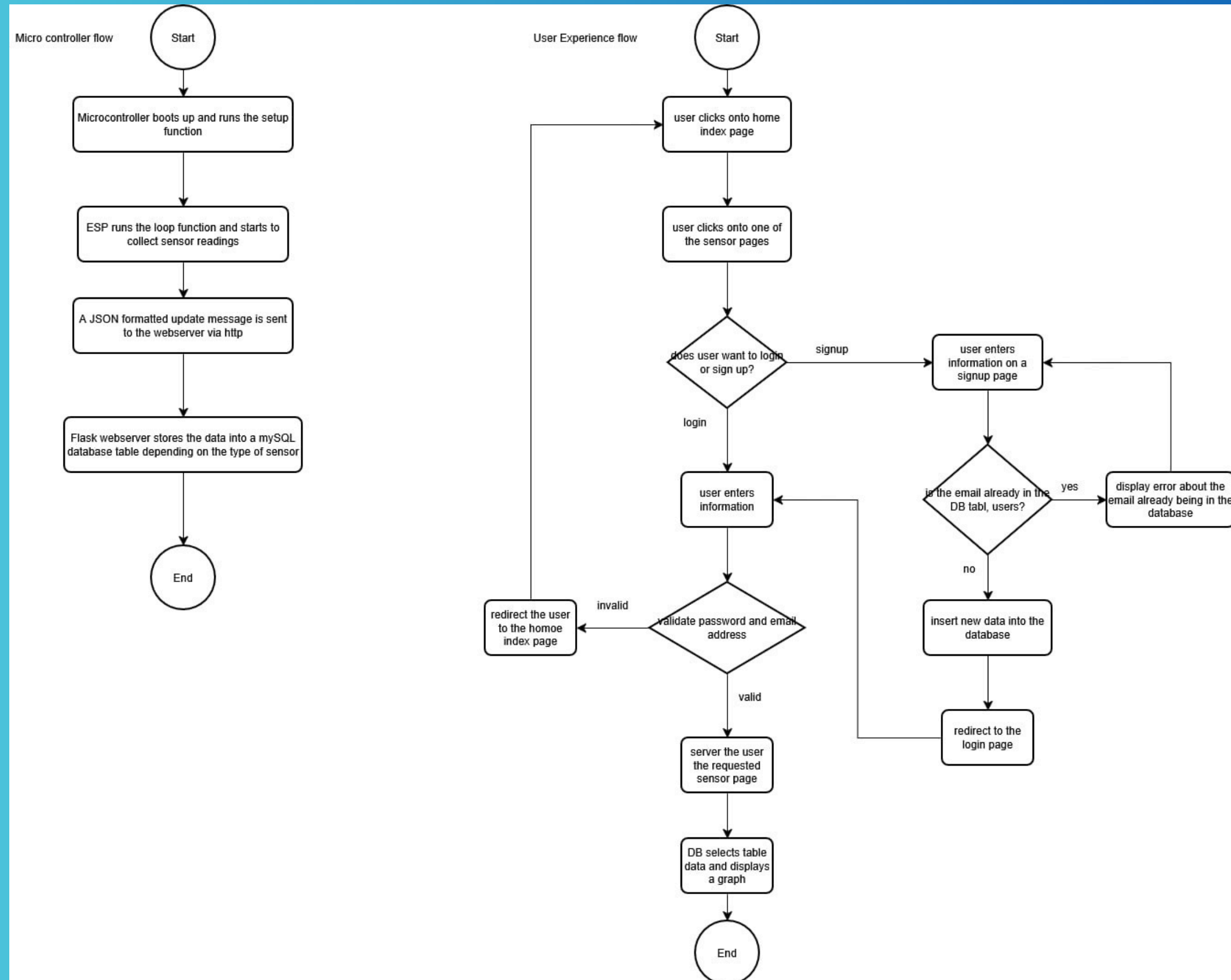
SYSTEM COMPONENTS



Components:

- Sensor/Microcontroller (Client)
- Wi-Fi network
- Flask Web server (Server)
- Database
- Client Browser

SYSTEM FLOW



HARDWARE

ULTRASONIC

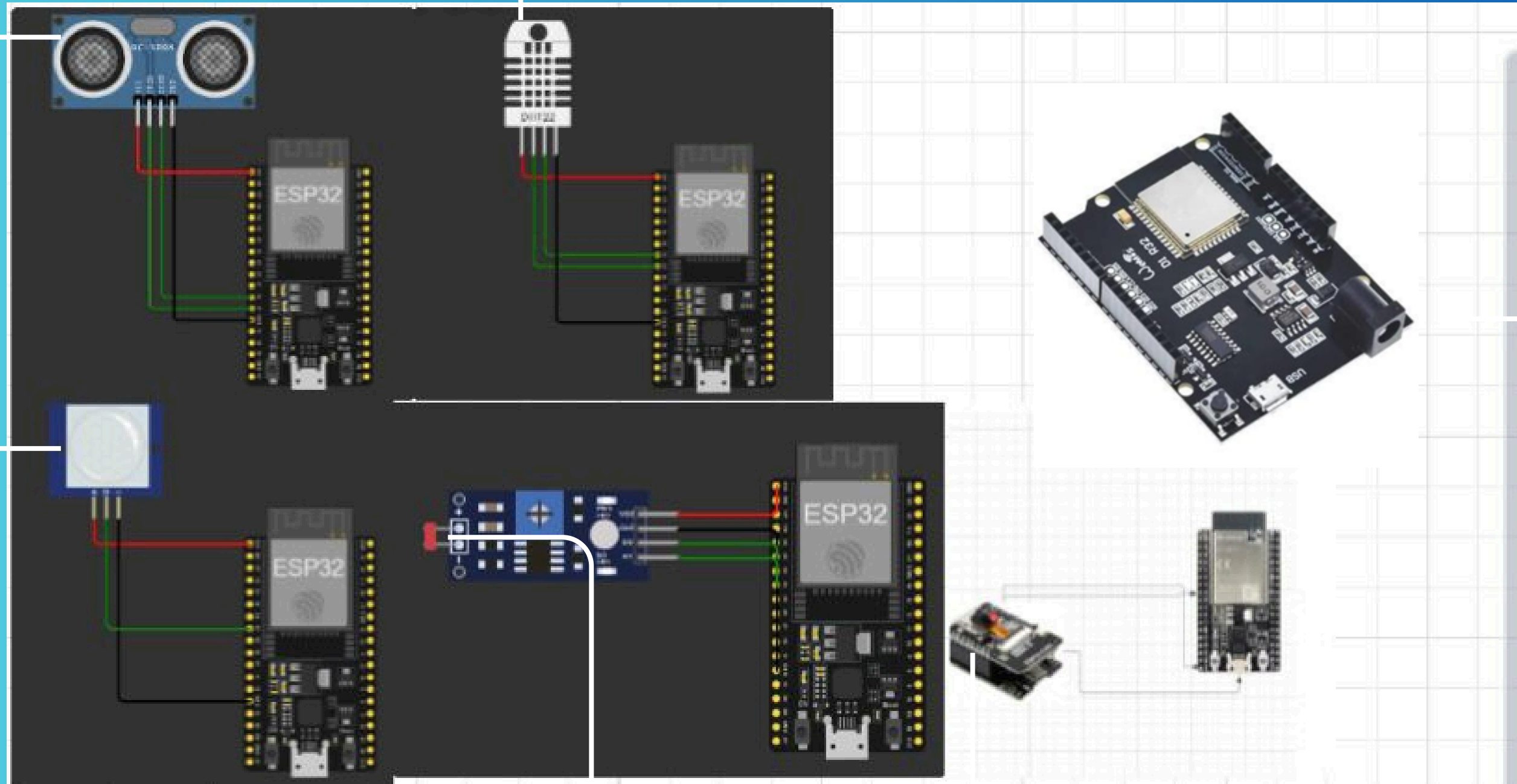
DHT22 SENSOR

ESP32 D1 R32

PIR SENSOR

LDR SENSOR

ESP32 CAMERA





LDR Sensor



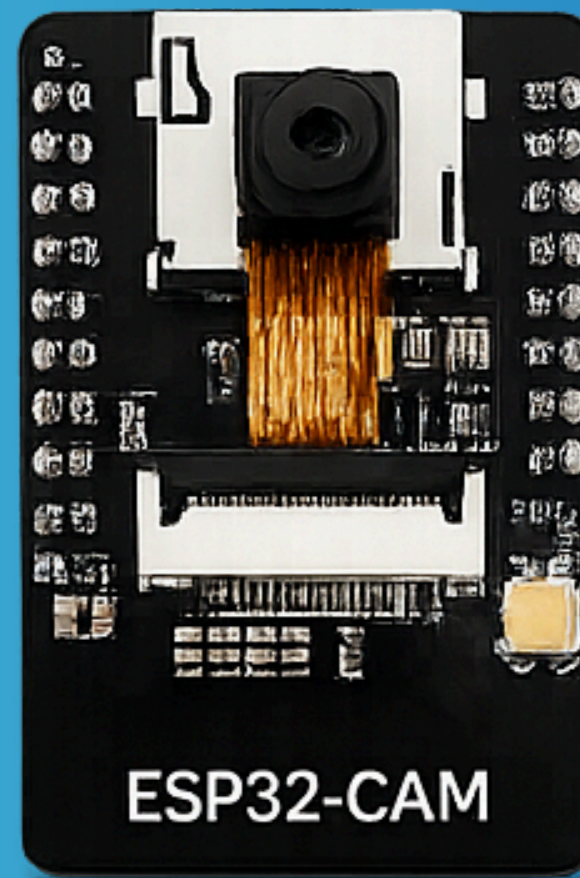
PIR Sensor (HC-SR505)



Ultrasonic Sensor (HC-SR04)



DHT22 Sensor



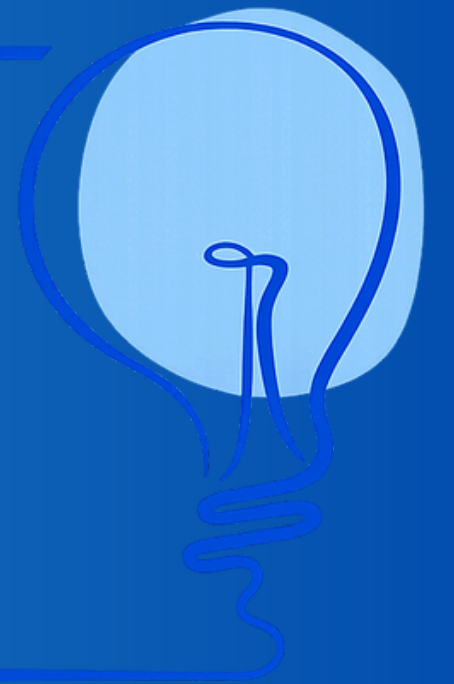
ESP32-CAM Module



ESP32 Microcontroller

- The smart home monitoring and automation system improves home security and environmental monitoring.
- The system integrates multiple sensors and modules for real-time data collection.
- The ESP32 microcontroller efficiently processes and transmits sensor data through WiFi communication.
- The web server and database enable reliable data management and storage.
- The web application allows users to monitor and control the system remotely.
- Real-time alerts and monitoring enhance safety and convenience for homeowners.
- The system demonstrates the effective use of IoT technology in smart home applications.

Conclusion



Thank You

For your attention!